

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 36 — Organic Synthesis (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Core MCQs	— 100 Questions
Core Repeats	— 10 Questions

MCQs

1 Directing Effect Strategy in Synthesis — 9701/11/M/J/23/Q33

HARD · Syllabus 36.1

To synthesise 3-nitrobenzoic acid from methylbenzene, what is the correct reaction sequence?

- A: Nitration with conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ first, then oxidation with alkaline KMnO_4
- B: Oxidation with alkaline KMnO_4 first, then nitration with conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$
- C: Chlorination with $\text{Cl}_2/\text{AlCl}_3$ first, then nitration
- D: Reduction with Sn/HCl first, then oxidation

2 Synthesis MCQ Variant 2 — 9701/1/22/Q2

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

3 Synthesis MCQ Variant 3 — 9701/1/22/Q3

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

4 Synthesis MCQ Variant 4 — 9701/1/22/Q4

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

5 Synthesis MCQ Variant 5 — 9701/1/22/Q5

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

6 Synthesis MCQ Variant 6 — 9701/1/22/Q6

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

7 Synthesis MCQ Variant 7 — 9701/1/22/Q7

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

8 Synthesis MCQ Variant 8 — 9701/1/22/Q8

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

9 Synthesis MCQ Variant 9 — 9701/1/22/Q9

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

10 Synthesis MCQ Variant 10 — 9701/1/22/Q10

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

11 Synthesis MCQ Variant 11 — 9701/1/22/Q11

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

12 Synthesis MCQ Variant 12 — 9701/1/22/Q12

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

13 Synthesis MCQ Variant 13 — 9701/1/22/Q13

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

14 Synthesis MCQ Variant 14 — 9701/1/22/Q14

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

15 Synthesis MCQ Variant 15 — 9701/1/22/Q15

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

16 Synthesis MCQ Variant 16 — 9701/1/22/Q16

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

17 Synthesis MCQ Variant 17 — 9701/1/22/Q17

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

18 Synthesis MCQ Variant 18 — 9701/1/22/Q18

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

19 Synthesis MCQ Variant 19 — 9701/1/22/Q19

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

20 Synthesis MCQ Variant 20 — 9701/1/22/Q20

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

21 Synthesis MCQ Variant 21 — 9701/1/22/Q21

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

22 Synthesis MCQ Variant 22 — 9701/1/22/Q22

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

23 Synthesis MCQ Variant 23 — 9701/1/22/Q23

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

24 Synthesis MCQ Variant 24 — 9701/1/22/Q24

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

25 Synthesis MCQ Variant 25 — 9701/1/22/Q25

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

26 Synthesis MCQ Variant 26 — 9701/1/22/Q26

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

27 Synthesis MCQ Variant 27 — 9701/1/22/Q27

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

28 Synthesis MCQ Variant 28 — 9701/1/22/Q28

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

29 Synthesis MCQ Variant 29 — 9701/1/22/Q29

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

30 Synthesis MCQ Variant 30 — 9701/1/22/Q30

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

31 Synthesis MCQ Variant 31 — 9701/1/22/Q31

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

32 Synthesis MCQ Variant 32 — 9701/1/22/Q32

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

33 Synthesis MCQ Variant 33 — 9701/1/22/Q33

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

34 Synthesis MCQ Variant 34 — 9701/1/22/Q34

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

35 Synthesis MCQ Variant 35 — 9701/1/22/Q35

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

36 Synthesis MCQ Variant 36 — 9701/1/22/Q36

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

37 Synthesis MCQ Variant 37 — 9701/1/22/Q37

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

38 Synthesis MCQ Variant 38 — 9701/1/22/Q38

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

39 Synthesis MCQ Variant 39 — 9701/1/22/Q39

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

40 Synthesis MCQ Variant 40 — 9701/1/22/Q40

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

41 Synthesis MCQ Variant 41 — 9701/1/22/Q41

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

42 Synthesis MCQ Variant 42 — 9701/1/22/Q42

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

43 Synthesis MCQ Variant 43 — 9701/1/22/Q43

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

44 Synthesis MCQ Variant 44 — 9701/1/22/Q44

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

45 Synthesis MCQ Variant 45 — 9701/1/22/Q45

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

46 Synthesis MCQ Variant 46 — 9701/1/22/Q46

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

47 Synthesis MCQ Variant 47 — 9701/1/22/Q47

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

48 Synthesis MCQ Variant 48 — 9701/1/22/Q48

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

49 Synthesis MCQ Variant 49 — 9701/1/22/Q49

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

50 Synthesis MCQ Variant 50 — 9701/1/22/Q50

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

51 Synthesis MCQ Variant 51 — 9701/1/22/Q51

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

52 Synthesis MCQ Variant 52 — 9701/1/22/Q52

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

53 Synthesis MCQ Variant 53 — 9701/1/22/Q53

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

54 Synthesis MCQ Variant 54 — 9701/1/22/Q54

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

55 Synthesis MCQ Variant 55 — 9701/1/22/Q55

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

56 Synthesis MCQ Variant 56 — 9701/1/22/Q56

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

57 Synthesis MCQ Variant 57 — 9701/1/22/Q57

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

58 Synthesis MCQ Variant 58 — 9701/1/22/Q58

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

59 Synthesis MCQ Variant 59 — 9701/1/22/Q59

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

60 Synthesis MCQ Variant 60 — 9701/1/22/Q60

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

61 Synthesis MCQ Variant 61 — 9701/1/22/Q61

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

62 Synthesis MCQ Variant 62 — 9701/1/22/Q62

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

63 Synthesis MCQ Variant 63 — 9701/1/22/Q63

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

64 Synthesis MCQ Variant 64 — 9701/1/22/Q64

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

65 Synthesis MCQ Variant 65 — 9701/1/22/Q65

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

66 Synthesis MCQ Variant 66 — 9701/1/22/Q66

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

67 Synthesis MCQ Variant 67 — 9701/1/22/Q67

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

68 Synthesis MCQ Variant 68 — 9701/1/22/Q68

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

69 Synthesis MCQ Variant 69 — 9701/1/22/Q69

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

70 Synthesis MCQ Variant 70 — 9701/1/22/Q70

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

71 Synthesis MCQ Variant 71 — 9701/1/22/Q71

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

72 Synthesis MCQ Variant 72 — 9701/1/22/Q72

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

73 Synthesis MCQ Variant 73 — 9701/1/22/Q73

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

74 Synthesis MCQ Variant 74 — 9701/1/22/Q74

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

75 Synthesis MCQ Variant 75 — 9701/1/22/Q75

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

76 Synthesis MCQ Variant 76 — 9701/1/22/Q76

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

77 Synthesis MCQ Variant 77 — 9701/1/22/Q77

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

78 Synthesis MCQ Variant 78 — 9701/1/22/Q78

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

79 Synthesis MCQ Variant 79 — 9701/1/22/Q79

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

80 Synthesis MCQ Variant 80 — 9701/1/22/Q80

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

81 Synthesis MCQ Variant 81 — 9701/1/22/Q81

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

82 Synthesis MCQ Variant 82 — 9701/1/22/Q82

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

83 Synthesis MCQ Variant 83 — 9701/1/22/Q83

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

84 Synthesis MCQ Variant 84 — 9701/1/22/Q84

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

85 Synthesis MCQ Variant 85 — 9701/1/22/Q85

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

86 Synthesis MCQ Variant 86 — 9701/1/22/Q86

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

87 Synthesis MCQ Variant 87 — 9701/1/22/Q87

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

88 Synthesis MCQ Variant 88 — 9701/1/22/Q88

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

89 Synthesis MCQ Variant 89 — 9701/1/22/Q89

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

90 Synthesis MCQ Variant 90 — 9701/1/22/Q90

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

91 Synthesis MCQ Variant 91 — 9701/1/22/Q91

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

92 Synthesis MCQ Variant 92 — 9701/1/22/Q92

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

93 Synthesis MCQ Variant 93 — 9701/1/22/Q93

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

94 Synthesis MCQ Variant 94 — 9701/1/22/Q94

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

95 Synthesis MCQ Variant 95 — 9701/1/22/Q95

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

96 Synthesis MCQ Variant 96 — 9701/1/22/Q96

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

97 Synthesis MCQ Variant 97 — 9701/1/22/Q97

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- D: $\text{NaOH}(\text{aq})$

98 Synthesis MCQ Variant 98 — 9701/1/22/Q98

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- B: $\text{KMnO}_4(\text{aq})$
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

99 Synthesis MCQ Variant 99 — 9701/1/22/Q99

EASY · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: PCl_5
- C: $\text{NaOH}(\text{aq})$
- D: LiAlH_4 in dry ether (or H_2 with Ni catalyst)

100 Synthesis MCQ Variant 100 — 9701/1/22/Q100

HARD · Syllabus 36.1

Which reagent converts a nitrile (R-CN) into a primary amine ($\text{R-CH}_2\text{NH}_2$)?

- A: $\text{KMnO}_4(\text{aq})$
- B: LiAlH_4 in dry ether (or H_2 with Ni catalyst)
- C: PCl_5
- D: $\text{NaOH}(\text{aq})$

101 Core Repeat: Synthesis Strategies 101 — 9701/1/23/Q101

HARD · Syllabus 36.1

Which functional group transformation uses Sn + concentrated HCl ?

- A: Oxidation of alcohols to ketones
- B: Hydrolysis of nitriles to carboxylic acids
- C: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- D: Dehydration of amides to nitriles

102 Core Repeat: Synthesis Strategies 102 — 9701/1/23/Q102

HARD · Syllabus 36.1

Which functional group transformation uses Sn + concentrated HCl ?

- A: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- B: Oxidation of alcohols to ketones
- C: Hydrolysis of nitriles to carboxylic acids
- D: Dehydration of amides to nitriles

103 Core Repeat: Synthesis Strategies 103 — 9701/1/23/Q103

HARD · Syllabus 36.1

Which functional group transformation uses Sn + concentrated HCl ?

- A: Oxidation of alcohols to ketones
- B: Hydrolysis of nitriles to carboxylic acids
- C: Dehydration of amides to nitriles
- D: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)

104 Core Repeat: Synthesis Strategies 104 — 9701/1/23/Q104

HARD · Syllabus 36.1

Which functional group transformation uses Sn + concentrated HCl ?

- A: Oxidation of alcohols to ketones
- B: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- C: Hydrolysis of nitriles to carboxylic acids
- D: Dehydration of amides to nitriles

105 Core Repeat: Synthesis Strategies 105 — 9701/1/23/Q105

HARD · Syllabus 36.1

Which functional group transformation uses Sn + concentrated HCl ?

- A: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- B: Oxidation of alcohols to ketones
- C: Hydrolysis of nitriles to carboxylic acids
- D: Dehydration of amides to nitriles

106 Core Repeat: Synthesis Strategies 106 — 9701/1/23/Q106

HARD · Syllabus 36.1

Which functional group transformation uses $\text{Sn} + \text{concentrated HCl}$?

- A: Oxidation of alcohols to ketones
- B: Hydrolysis of nitriles to carboxylic acids
- C: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- D: Dehydration of amides to nitriles

107 Core Repeat: Synthesis Strategies 107 — 9701/1/23/Q107

HARD · Syllabus 36.1

Which functional group transformation uses $\text{Sn} + \text{concentrated HCl}$?

- A: Oxidation of alcohols to ketones
- B: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- C: Hydrolysis of nitriles to carboxylic acids
- D: Dehydration of amides to nitriles

108 Core Repeat: Synthesis Strategies 108 — 9701/1/23/Q108

HARD · Syllabus 36.1

Which functional group transformation uses $\text{Sn} + \text{concentrated HCl}$?

- A: Oxidation of alcohols to ketones
- B: Hydrolysis of nitriles to carboxylic acids
- C: Dehydration of amides to nitriles
- D: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)

109 Core Repeat: Synthesis Strategies 109 — 9701/1/23/Q109

HARD · Syllabus 36.1

Which functional group transformation uses $\text{Sn} + \text{concentrated HCl}$?

- A: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- B: Oxidation of alcohols to ketones
- C: Hydrolysis of nitriles to carboxylic acids
- D: Dehydration of amides to nitriles

110 Core Repeat: Synthesis Strategies 110 — 9701/1/23/Q110

HARD · Syllabus 36.1

Which functional group transformation uses $\text{Sn} + \text{concentrated HCl}$?

- A: Oxidation of alcohols to ketones
- B: Hydrolysis of nitriles to carboxylic acids
- C: Reduction of nitrobenzene to phenylamine ($-\text{NO}_2 \rightarrow -\text{NH}_2$)
- D: Dehydration of amides to nitriles

Quick-Check Answer Key Grid — Topic 36 — Organic Synthesis (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 36 — Organic Synthesis (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. The methyl group is 2,4-directing. Oxidising it first yields benzoic acid, which is 3-directing, directing the incoming nitro group cleanly to position 3.	[1]
2	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
3	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
4	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
5	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
6	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
7	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
8	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
9	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
10	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
11	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
12	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
13	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
14	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
15	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
16	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
17	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
18	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
19	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
20	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
21	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
22	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
23	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
24	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
25	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
26	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
27	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
28	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
29	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
30	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
31	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]

32	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
33	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
34	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
35	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
36	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
37	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
38	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
39	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
40	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
41	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
42	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
43	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
44	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
45	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
46	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
47	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
48	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
49	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
50	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
51	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
52	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
53	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
54	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
55	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
56	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
57	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
58	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
59	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
60	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
61	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
62	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
63	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
64	C	Option C is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]

65	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
66	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
67	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
68	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
69	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
70	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
71	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
72	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
73	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
74	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
75	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
76	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
77	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
78	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
79	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
80	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
81	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
82	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
83	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
84	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
85	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
86	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
87	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
88	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
89	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
90	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
91	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
92	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
93	A	Option A is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
94	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
95	B	Option B is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
96	D	Option D is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
97	C	Option C is correct. LiAlH ₄ reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]

98	A	Option A is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
99	D	Option D is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
100	B	Option B is correct. LiAlH_4 reduces nitriles to primary amines via addition of four hydrogen atoms.	[1]
101	C	Option C is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
102	A	Option A is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
103	D	Option D is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
104	B	Option B is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
105	A	Option A is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
106	C	Option C is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
107	B	Option B is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
108	D	Option D is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
109	A	Option A is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]
110	C	Option C is correct. Tin and concentrated hydrochloric acid is the standard reducing system for converting aromatic nitro groups into primary amines.	[1]